FIITJEE

JEE(Main)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

CONCEPT RECAPITULATION TEST – I

(Main)

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**B**	31.	**B**	61.	**A**
2.	**C**	32.	**B**	62.	**B**
3.	**C**	33.	**B**	63.	**C**
4.	**D**	34.	**C**	64.	**A**
5.	**A**	35.	**D**	65.	**C**
6.	**B**	36.	**B**	66.	**C**
7.	**D**	37.	**B**	67.	**C**
8.	**A**	38.	**D**	68.	**A**
9.	**B**	39.	**C**	69.	**B**
10.	**A**	40.	**C**	70.	**B**
11.	**B**	41.	**D**	71.	**D**
12.	**B**	42.	**C**	72.	**B**
13.	**C**	43.	**C**	73.	**A**
14.	**D**	44.	**B**	74.	**A**
15.	**C**	45.	**C**	75.	**A**
16.	**C**	46.	**B**	76.	**B**
17.	**C**	47.	**B**	77.	**D**
18.	**A**	48.	**C**	78.	**D**
19.	**B**	49.	**B**	79.	**D**
20.	**D**	50.	**C**	80.	**A**
21.	**D**	51.	**B**	81.	**B**
22.	**A**	52.	**B**	82.	**C**
23.	**D**	53.	**C**	83.	**B**
24.	**C**	54.	**B**	84.	**B**
25.	**D**	55.	**B**	85.	**C**
26.	**C**	56.	**B**	86.	**B**
27.	**A**	57.	**C**	87.	**B**
28.	**D**	58.	**C**	88.	**B**
29.	**A**	59.	**A**	89.	**C**
30.	**B**	60.	**B**	90.	**C**

Physics

PART – I

SECTION – A

1. B

Sol. capillary rise

$$h = \frac{2T\cos(\theta)}{r\rho g}$$

$$\therefore \frac{h_1}{h_2} = \frac{r_2}{r_1} = \frac{0.4}{0.2} = 2:1$$

2. C

Sol. $x = A\sin(\omega t + \phi_o)$

At t = 0, x = 0 $\Rightarrow A\sin\phi_o = 0$ or $\phi_o = 0$

Hence, $x = A\sin(\omega t)$

or A/2 = A sin (ωt)

or 1/2 = sin (ωt)

$$\omega t = \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

$$t = \frac{\pi}{6\omega} = \frac{\pi.T}{6(2\pi)}$$

as $\omega = 2\pi / T \Rightarrow$ t T/12 = 1/3 s

3. C

Sol. $$i = \frac{E}{R}\left(1 - e^{-\frac{tR}{L}}\right)$$

4. D

Sol. $$\frac{E_{4n} - E_{2n}}{E_{2n} - E_n} = \frac{\frac{E_1}{16n^2} - \frac{E_1}{4n^2}}{\frac{E_1}{4n^2} - \frac{E_1}{n^2}} = \frac{1}{4} = \text{constant}$$

5. A

Sol. $f\lambda = v = \frac{10\lambda v}{d} \Rightarrow \lambda = \frac{d}{10}$.

Maximum path diff. can be $\frac{d}{3}$. Hence, number of maximum will be 6.

6. B

Sol. $C_{rms} = \sqrt{\frac{3RT}{M}}$ or $C_{rms} \alpha \sqrt{T}$

$$\frac{(C_{rms})_{T_2}}{(C_{rms})_{T_1}} = \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{400}{100}} = 2$$

$(C_{rms})_{T_2} = 2(C_{rms})_{T_1} = 2 \times v = 2v$

7. D

Sol. $t = 20 \text{ min} = \frac{1}{3}\text{hr}$

$1 \text{ km} = \sqrt{(6)^2 - V^2} \times \left(\frac{1}{3}\text{hr}\right)$

$V = 3\sqrt{3}$ km/hr.

8. A

Sol. $12.75 = \left|E_0 - \frac{E_0}{n^2}\right| = \left|-13.6 - \frac{(-13.6)}{n_1^2}\right|$

$\Rightarrow \quad n = 4$

no. of lines $= \frac{n(n-1)}{2} = 6$

9. B

Sol. $qvB = \frac{mv^2}{R}$

$mv = qRB = 1.6 \times 10^{-21}$ kg m/sec

10. A

Sol: Consider the differential element of the cylinder as shown in the figure.

$\therefore \qquad R = \int_{R_1}^{R_2} \rho \frac{dx}{2\pi x l} \qquad (\because R = \rho \frac{l}{a})$

$\Rightarrow R = \frac{\rho}{2\pi l} \ln\left(R_2 / R_1\right)$

$I = \frac{\varepsilon}{R}$; $\Rightarrow I = \frac{2\pi l \varepsilon}{\rho \ln\left(\frac{R_2}{R_1}\right)}$

11. B

Sol. +ve slope = +ve velocity and convex graph = –ve acceleration
Concave graph = +ve acceleration

12. B

Sol. $\frac{1}{2}mv^2$ = Power t = Fv t = kt

$\therefore \quad V = \sqrt{\frac{2kt}{m}}$

13. C

Sol: Let dq be the charge which has passed in a small interval of time dt, then dq = idt = (4 + 2t)dt. Hence total charge passed between interval t = 2 sec and t = 6 sec

$q = \int_2^6 (4+2t)dt$ = 48 coulomb.

14. D

Sol. Maximum stress on the wire will be at highest point (at the point of suspension)

$\therefore$ stress = $\frac{\text{weight}}{A} = \frac{Al\rho g}{A} = \sigma$

$\therefore$ $l = \frac{\sigma}{\rho g}$

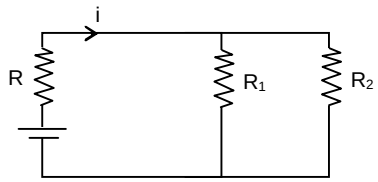
15. C

Sol. $T + f = mg \sin\theta$ (Translational equilibrium)

$T(R) = f(R)$ (Rotational equilibrium)

16. C

Sol. Inductor behaves like a cell of emf, $\varepsilon' = Blv$ and equivalent ckt is



$Blv = \varepsilon'$

Hence, $i = \frac{Blv}{R + \frac{R_1R_2}{R_1+R_2}}$

$= \frac{Blv(R_1+R_2)}{RR_1 + RR_2 + R_1R_2}$

17. C

Sol. $E = -\frac{\Delta V}{\Delta r}$ directed $\perp^r$ to an equipotential surface.

18. A

Sol. Inside water

u = –20 cm

f = –15 cm

$\Rightarrow$ v = –60 cm

$m = \frac{-v}{u}$ = –3

$\Rightarrow$ n = 3

19. B

Sol. $C = C_v + \frac{PdV}{ndT}$,

$PV^{3/2} = k$, $PV = nRT$

find relation in V and T

$\frac{dV}{dT} = -\frac{2V}{T}$

$\therefore$ $C = C_v - 2R$

20. D

Sol. $A_1V_1 = A_2V_2 \Rightarrow a\sqrt{2gh} = A\left(-\frac{dh}{dt}\right)$

$\Rightarrow \quad \frac{a}{A}\int_0^t dt = -A\int_{h_1}^{h_2}\frac{dh}{\sqrt{2gh}}$

21. D

Sol. No relative motion between source and observers.

22. A

Sol. Heat energy flowing per sec is given by

$H = \frac{Q}{t} = KA\frac{\Delta\theta}{\Delta x}$ . . . (i)

if dm mass of ice is increased in time dt, then

$\frac{dm}{dt} = \frac{A\,dx\,\rho}{dt} = A.\rho.\frac{dx}{dt}$

Since, $H = \left(\frac{dm}{dt}\right)L$

$\therefore \quad H = A\rho\frac{dx}{dt}L$ . . . (ii)

From eq. (I) and (II)

$A\rho L\frac{dx}{dt} = KA\frac{\Delta\theta}{\Delta x}$

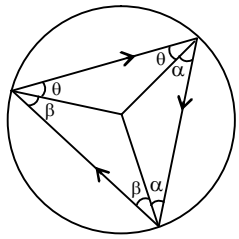
Rate of thickening of ice = dx/dt

$\therefore \quad \frac{dx}{dt} = \frac{KA}{\rho AL}\frac{\Delta\theta}{\Delta x} = \frac{K}{\rho L}\frac{\Delta\theta}{\Delta x} = \frac{4\times10^{-4}}{0.9\times10^3\times80}\times\left[\frac{0-(-15)}{2\times10^{-2}}\right]$

$= 4.166\times10^{-6}$ m/s = 1.5 cm/ hour.

23. D

Sol. $\tan\alpha = \frac{u\sin\theta}{eu\cos\theta} = \frac{\tan\theta}{e}$



$\therefore \quad \tan\alpha = \frac{1}{e}\tan\theta$

$\therefore \quad \tan\beta = \frac{1}{e}\tan\theta$

$2\theta + 2\alpha + 2\beta = 180°$

$\alpha + \beta = (90 - \theta)$

$\tan(\alpha + \beta) = \cot\theta$

$\Rightarrow \quad \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta} = \frac{1}{\tan\theta}$

$\Rightarrow \quad \frac{\frac{1}{e}\tan\theta + \frac{1}{e^2}\tan\theta}{1 - \frac{1}{e^3}\tan^2\theta} = \frac{1}{\tan\theta}$

$$\Rightarrow \tan^2\theta\left(\frac{1}{e}+\frac{1}{e^2}\right)=1-\frac{1}{e^3}\tan^2\theta$$

$$\Rightarrow \tan^2\theta\left(\frac{1}{e}+\frac{1}{e^2}+\frac{1}{e^3}\right)=1$$

$$\tan^2\theta\left(\frac{1}{\frac{1}{e}+\frac{1}{e^2}+\frac{1}{e^3}}\right)=\frac{e^3}{1+e+e^2}$$

24. C

Sol. Let the cylinder is slightly displaced from mean position.
The total energy of the system when elongation in spring is x.

$$E=\frac{1}{2}kx^2+\frac{1}{2}Mv^2+\frac{1}{2}\times\left(\frac{1}{2}MR^2\right)\omega^2$$

Differentiating the above expression, we get

$$2kx\frac{dx}{dt}+M2v\frac{dv}{dt}+\frac{1}{2}MR^2\frac{2v}{R^2}\frac{dv}{dt}=0$$

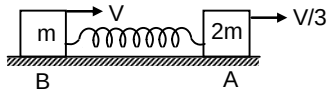
$$2kx+(2M+M)\frac{dv}{dt}=0$$

$$\frac{dv}{dt}=-\left(\frac{2K}{3M}\right)x$$

$$T=2\pi\sqrt{\frac{3M}{2k}}$$

25. D

Sol. Velocity of A after the collision with C, $V_1=\frac{V}{3}$



$\therefore$ Velocity of each block A and B at maximum compression

$$V_2=\frac{5}{9}V$$

$$\therefore \frac{1}{2}k\,x^2+\frac{1}{2}3m\left(\frac{5}{9}V\right)^2$$

$$=\frac{1}{2}mv^2+\frac{1}{2}2m\left(\frac{V}{3}\right)^2$$

$$x=\sqrt{\frac{8m}{27K}}V$$

26. C

Sol. Velocity of ball 'A' just after the collision $V=\left(1+\frac{1}{2}\right)\cdot\frac{2}{3}u=u$

Velocity of ball 'B' just after the collision = 0.

$$\therefore \omega=\frac{u-0}{L}=\frac{u}{L}$$

27. A

Sol. $T = \frac{t}{20}$°C where t is time in seconds.

$$\frac{d\theta}{dt} = \frac{kA(T-0)}{\ell}$$

$$\Delta Q = \int dQ = \frac{kA}{\ell}\int \frac{t}{20} dt = 18000 \text{ J.}$$

28. D

Sol. $v = \sqrt{\frac{T}{\mu}}$; $V_{AB} = \sqrt{\frac{T}{\mu}}$; $V_{CD} = \sqrt{\frac{2T}{\mu}}$

29. A

Sol. $\frac{4S}{R_1} = 4\left(\frac{4S}{R_2}\right)$

$\Rightarrow \quad R_2 = 4R_1$

Required ratio = $\frac{\frac{4}{3}\pi R_1^3}{\frac{4}{3}\pi R_2^3}$

30. B

Sol. At the level of interface

$P_{left} = P_{right}$

$\rho_1 g h_1 = \rho_2 g h_2$

$\Rightarrow \quad \rho \propto \frac{1}{h}$.

Chemistry PART – II

SECTION – A

31. B
Sol. In H-atom, the orbitals(five 3d, three 3p and one 3s) have same energy. So, they are called degenerate orbitals.

32. B
Sol. It is a zero order reaction.

33. B
Sol. Increasing the volume of reaction container twice, the concentration of all the species will reduce to half.

34. C
Sol. $pH = p_{Ka} + \log\frac{[CH_3COONa]}{[CH_3COOH]} = 5 + \log\frac{200\times 0.4}{400\times 0.2} = 5$

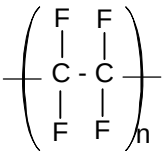
35. D
Sol. The observed bond angles are 120°, 180° and 90°.

36. B
Sol. $4LiNO_3 \longrightarrow 2Li_2O + 4NO_2 + O_2$

37. B
Sol. Peroxoborates contain peroxide linkages, which behave as bleaching agent.

38. D
Sol. In chlorobenzene, Cl exerts –I and +R effect. In other compounds, it only exerts –I effect.

39. C
Sol. Carbon should be present in the compound for giving nitrogen test.

40. C
Sol. Teflon is 

41. D
Sol. $Ca_3P_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3$

42. C
Sol. $NaCl + H_2SO_4 \longrightarrow NaHSO_4 + HCl$

$2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2$

$2HCl + O_3 \longrightarrow Cl_2 + O_2 + H_2O$

$KCl + AgNO_3 \longrightarrow AgCl + KNO_3$

43. C

Sol. Due to stable $t_{2g}^3 e_g^2$ configuration of Fe^{3+} ion.

44. B

Sol. Roasting is carried out for sulphide ores.

45. C

Sol. It produces maximum number of ions as compared to other salts.

46. B

Sol. The sol becomes positively charged due to excess of Fe^{3+} ions.

47. B

Sol. $E = E^o - \frac{0.0591}{n}\log\left[H^+\right] = 0 - \frac{0.0591}{1}\log 10^{-4} = 0.236\,V$

48. C

Sol. $XY_3(g) \longrightarrow X(g) + 3Y(g)$; ΔH = E kJ mol^{-1}

$\therefore$ B.E of XY = $\frac{E}{3}$ as XY_3 contains three X – Y bonds.

49. B

Sol. For spontaneous process, $\Delta G < 0$.

$\therefore \Delta H - T\Delta S < 0$, $\Delta H < T\Delta S$

$\therefore\ T > \frac{\Delta H}{\Delta S} = \frac{18000}{60} = 300K$

50. C

Sol. Moles of HCl = $\frac{1200 \times 0.4}{1000} = 0.48$

Moles of NaOH = $\frac{19.2}{40} = 0.48$

51. B

Sol. Easily liquefiable gases should have high critical temperature(T_C), which is given by

$T_C = \frac{8a}{27Rb} = \left(\frac{8}{27R}\right)\frac{a}{b}$

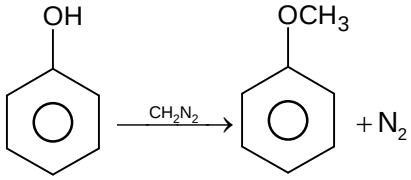
$\therefore T_C\ \alpha \frac{a}{b}$

52. B
Sol. $X = Na_2SO_3$
$Y = BaSO_3$
$Z = BaSO_4$

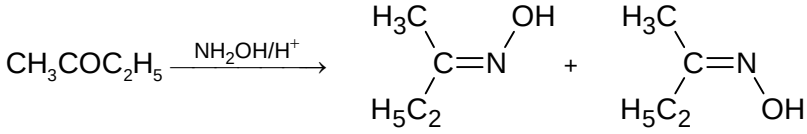
53. C
Sol. CsCl has b.c.c. unit cell.

54. B
Sol. 'CO' can form carboxy haemoglobin.

55. B
Sol. Sucrolose is an artificial sweetner.

56. B
Sol. 

57. C
Sol. Anti Markownikoff reaction takes place in this case.

58. C
Sol. 

59. A
Sol. $CH_3CH_2CH_2CONH_2 \xrightarrow[KOH]{Br_2} CH_3CH_2CH_2NH_2$

60. B
Sol. Due to intermolecular hydrogen bonding.

Mathematics PART – III

SECTION – A

61. A

Sol. $\because \quad S=\sum_{r=1}^{16}\frac{8r}{\left(4r^4+1\right)}$

$$=\sum_{r=1}^{16}\frac{8r}{\left(2r^2-2r+1\right)\left(2r^2+2r+1\right)}$$

$$=2\sum_{r=1}^{16}\left(\frac{1}{2r^2-2r+1}-\frac{1}{2r^2+2r+1}\right)$$

$$=2\left\{\frac{1}{1}-\frac{1}{5}+\frac{1}{5}-\frac{1}{13}+\frac{1}{13}+.....+\frac{1}{481}-\frac{1}{545}\right\} \quad =2\left(1-\frac{1}{545}\right)=\frac{1088}{545}$$

62. B

Sol. $f'(x)-\frac{2x(x+1)}{x+1}f(x)=\frac{e^{x^2}}{(x+1)^2}$

$$IF = e^{\int -2x\,dx}=e^{-x^2}$$

$$\Rightarrow \quad f(x).\left(-e^{-x^2}\right)=\int\frac{dx}{(x+1)^2}$$

$$\Rightarrow \quad f(x).e^{-x^2}=\frac{1}{x+1}+C$$

At x = 0, f(0) = 5

$$\Rightarrow \quad C = 6$$

$$\therefore \quad f(x)=\left(\frac{6x+5}{x+1}\right).e^{x^2}$$

Hence, (b) is the correct answer.

63. C

Sol. We have $\vec{r}\times\vec{a}=\vec{b}\times\vec{a}$

$$\Rightarrow \quad \left(\vec{r}-\vec{b}\right)\times\vec{a}=0$$

$$\Rightarrow \quad \vec{r}-\vec{b} \text{ is parallel to } \vec{a}$$

$$\Rightarrow \quad \vec{r}=\vec{b}+\lambda\vec{a} \qquad \ldots\ldots\ldots\ldots\text{(i)}$$

Now, $\vec{r}.\vec{c}=0$

$$\Rightarrow \quad \left(\vec{b}+\lambda\vec{a}\right).\vec{c}=0$$

$$\Rightarrow \quad \vec{b}\,.\,\vec{c}+\lambda\left(\vec{a}.\vec{c}\right)=0$$

$$\Rightarrow \quad \lambda=-\frac{\vec{b}.\vec{c}}{\vec{a}.\vec{c}}$$

On putting the value of λ in Eq. (i), we get

$$\vec{r} = \vec{b} - \frac{\left(\vec{b}.\vec{c}\right)}{\left(\vec{a}.\vec{c}\right)}\vec{a}$$

Hence, (c) is the correct answer.

64. A

Sol. Replacing f (x) by x, we f (x) = x^2

$\Rightarrow$ ffff (x) = x^{16} also $(f(x^4))^2 = x^{16}$.

65. C

Sol. Since coefficient of $x^2 + ax + 1$ are symmetrical so roots will be α, $\frac{1}{\alpha}$

$$\lim_{x\to 1/\alpha} \frac{\sin\left(x^2+ax+1\right)}{(\alpha x-1)} = \lim_{x\to 1/\alpha} \frac{\sin\left[\left(x-\frac{1}{\alpha}\right)(x-\alpha)\right]}{\left(x-\frac{1}{\alpha}\right)(x-\alpha)} \cdot \frac{\left(x-\frac{1}{\alpha}\right)(x-\alpha)}{(\alpha x-1)}$$

$$= \lim_{x\to 1/\alpha} \frac{(x-\alpha)}{\alpha} = \frac{\frac{1}{\alpha}-\alpha}{\alpha} = \frac{1-\alpha^2}{\alpha^2}$$

66. C

Sol. If u = f(cos x) and v = g (sinx)

$$\Rightarrow \quad \frac{du}{dv} = \frac{-f'(\cos x)\sin x}{g'(\sin x)\cos x}$$

At $x = \frac{\pi}{3}$, $= \frac{-f'\left(\frac{1}{2}\right)\tan\frac{\pi}{3}}{g'\left(\frac{\sqrt{3}}{2}\right)} = \frac{-2}{-2}\times\sqrt{3} = \sqrt{3}$

67. C

Sol. Given that f (x) = |1 – x|

$$\Rightarrow f(|x|) = \begin{cases} x-1, & x>1 \\ 1-x, & 0<x\le 1 \\ 1+x, & -1\le x\le 0 \\ -x-1, & x<-1 \end{cases}$$

Clearly the domain of $\sin^{-1}$ (f |x|) is [–2, 2].

$\Rightarrow$ It is non differentiable at the points {– 1, 0, 1}.

68. A

Sol. y = x touches $y = x^2 + bx + c$

$1 = 1 + b + c \Rightarrow b + c = 0$ ……(1)

$$\frac{dy}{dx} = 2x + b\big|_{(1,\,1)} = 2 + b = 1 \Rightarrow b = -1,\ c = 1.$$

(b, c) = (–1, 1).

69. B

Sol. $\int e^x\left(\frac{2\tan x}{1+\tan x}+\tan^2\left(x-\frac{\pi}{4}\right)\right)dx$

$=\int e^x\left(\tan\left(x-\frac{\pi}{4}\right)+\sec^2\left(x-\frac{\pi}{4}\right)\right)dx$

$=e^x\tan\left(x-\frac{\pi}{4}\right)+c$.

70. B

Sol. $\int\frac{dx}{(\sqrt{x})^2+(\sqrt{x})^7}=\int\frac{dx}{(\sqrt{x})^7\left(1+\frac{1}{(\sqrt{x})^5}\right)}$

Put $\frac{1}{(\sqrt{x})^5}=y$

$\frac{dy}{dx}=-\frac{5}{2(\sqrt{x})^7}$

$\therefore I=\int\frac{-2dy}{5(1+y)}=-\frac{2}{5}\ln|1+y|+c=+\frac{2}{5}\ln\left(\frac{1}{1+\frac{1}{(\sqrt{x})^5}}\right)$

$\therefore a=\frac{2}{5}, k=\frac{5}{2}$.

71 D

Sol. $\int_0^{\pi/4}\ln\left(\cos\left(\frac{\pi}{4}+x\right)\right)^{\cot\left(\frac{\pi}{4}-x\right)}dx=\int_0^{\pi/4}\cot\left(\frac{\pi}{4}-x\right).\ln\cos\left(\frac{\pi}{4}+x\right)dx$

$\int_0^{\pi/4}\cot\left(\frac{\pi}{4}-\left(\frac{\pi}{4}+0-x\right)\right)\ln\cos\left(\frac{\pi}{4}+\left(\frac{\pi}{4}+0-x\right)\right)dx$

$=\int_0^{\pi/4}\cot x.\ln(\sin x)dx=\frac{1}{2}(\ln\sin x)^2\Big|_0^{\pi/4}=$ not defined.

72. B

Sol. $\int_{-10}^{10}\frac{3^x}{3^{[x]}}dx=20\int_0^1 3^{x-[x]}dx=20\int_0^1 3^{\{x\}}dx=20\int_0^1 3^x dx$

$=20\left[\frac{3^x}{\ln 3}\right]_0^1=\frac{40}{\ln 3}$.

73. A

Sol. The distance between (2, 2m) and (5, 5m) is less then 5.

$\Rightarrow (5-2)^2+(5m-2m)^2<25$

$\Rightarrow 9m^2 < 16$

$m^2 < \left(\frac{4}{3}\right)^2 \Rightarrow -\frac{4}{3} < m < \frac{4}{3}$.

74. A

Sol. Let A denote the event that a sum of 5 occurs, B the event that a sum of 7 occurs and C the event that neither a sum of 5 nor a sum of 7 occurs. We have

$P(A) = \frac{4}{36} = \frac{1}{9}, P(B) = \frac{6}{36} = \frac{1}{6}$

$P(C) = \frac{26}{36} = \frac{13}{18}$

Thus, probability that A occurs before B is

$P\left[A \text{ or } (C \cap A) \text{or} (C \cap C \cap A) \text{or} ....\right]$

$= P(A) + P(C \cap A) + P(C \cap C \cap A) + ...$

$= P(A) + P(C)P(A) + P(C)^2 P(A) + ...$

$= \frac{1}{9} + \left(\frac{13}{18}\right) \times \frac{1}{9} + \left(\frac{13}{18}\right)^2 \frac{1}{9} + ...$

$= \frac{1/9}{1 - 13/18} = \frac{2}{5}$

75. A

Sol $\sum_{r=1}^{n+1} \left(\sum_{k=1}^{n} {}^{k}C_{r-1} \right)$

$= \sum_{r=1}^{n+1} \left(\sum_{k=1}^{n} \left({}^{k+1}C_r - {}^{k}C_r \right) \right)$

$= \sum_{r=1}^{n+1} \left({}^{n+1}C_r - {}^{1}C_r \right) = 2^{n+1} - 2$

76. B

Sol. $b_2 = \frac{1}{1 - b_1}$

$b_3 = \frac{1}{1 - b_2} = \frac{1}{1 - \frac{1}{1 - b_1}} = \frac{1 - b_1}{-b_1} = \frac{b_1 - 1}{b_1}$

$b_1 = b_3 \Rightarrow b_1^2 - b_1 + 1 = 0$

$\Rightarrow b_1 = -\omega \quad \text{or} \quad \omega^2 \Rightarrow b_2 = \frac{1}{1 + \omega} = -\omega \text{ or } \omega^2$

$\sum_{r=1}^{2001} {b_r}^{2001} = \sum_{r=1}^{2001} (-\omega)^{2001}$

$= \sum_{r=1}^{2001} -1 = -2001$

77. D

Sol. Let the plane $\vec{r}.\left(\vec{i}-\vec{2}j+3\vec{k}\right)=17$ divide the line joining the points $-2\vec{i}+\vec{4}j+7\vec{k}$ and $\vec{3}i--5\vec{j}+8\vec{k}$ in the ratio t : 1 at point P.

Therefore, point P is

$$\frac{3t-2}{t+1}\vec{i}+\frac{-5t+4}{t+1}\vec{j}+\frac{8t+7}{t+1}\vec{k}$$

This lies on the given plane

$$\therefore \frac{3t-2}{t+1}(1)+\frac{-5t+4}{t+1}(-2)+\frac{8t+7}{t+1}(3)=17$$

Solving, we get

$$t=\frac{3}{10}$$

78. D

Sol. Vector perpendicular to the face OAB is

$$\overrightarrow{OA}\times\overrightarrow{OB}=\left(\hat{i}+2\hat{j}+\hat{k}\right)\times\left(2\hat{i}+\hat{j}+3\hat{k}\right)$$

$$=5\hat{i}-\hat{j}-3\hat{k}$$

Vector perpendicular to face ABC is

$$\overrightarrow{AB}\times\overrightarrow{AC}=\left(\hat{i}-\hat{j}+2\hat{k}\right)\times\left(-2\hat{i}-\hat{j}+\hat{k}\right)$$

$$=\hat{i}-5\hat{j}-3\hat{k}$$

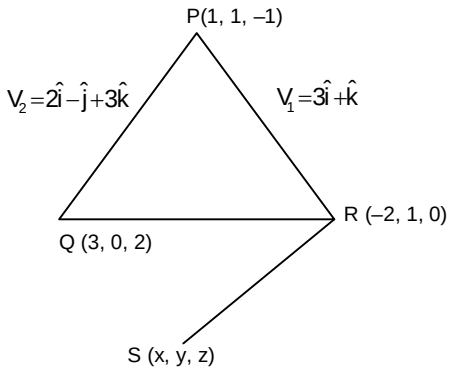
Since the angle between the face = angle between their normal, therefore

$$\cos\theta=\frac{5+5+9}{\sqrt{35}\sqrt{35}}=\frac{19}{35}\Rightarrow\theta=\cos^{-1}\left(\frac{19}{35}\right)$$

79. D

Sol. $\vec{V}_1, \vec{V}_2, \overrightarrow{PS}$ are in the same plane

$$\therefore \left(2\hat{i}-\hat{j}+3\hat{k}\right)\times\left(-3\hat{i}+\hat{k}\right).\left((x+2)\hat{i}+(y-1)\hat{j}+z\hat{k}\right)=0$$



80. A

Sol. $$\frac{\sin(xy)}{\cos(xy)}=y$$

$$\Rightarrow \sin(xy)=xy$$

$$\Rightarrow xy=0$$

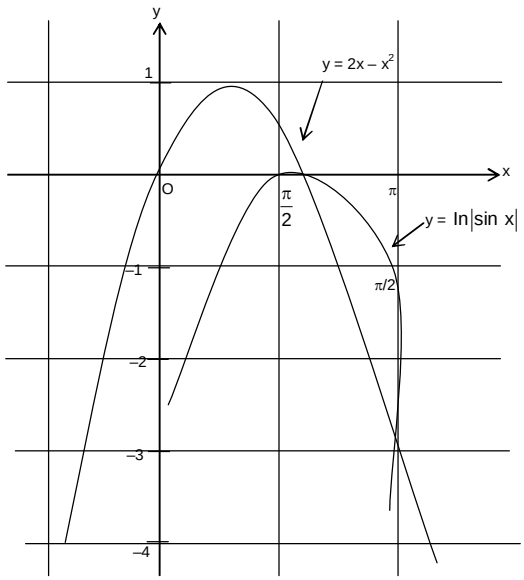
$\Rightarrow x=0$ or y = 0

But x = 0 is not possible

$\therefore$ y = 0 and x = 1, i.e. (1, 0)

81. B

Sol.



$\ln|\sin x| = -x(x-2)$

Graphs of $y = \ln|\sin x|$ and $y = -x(x-2)$ meet exactly two times in $\left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$

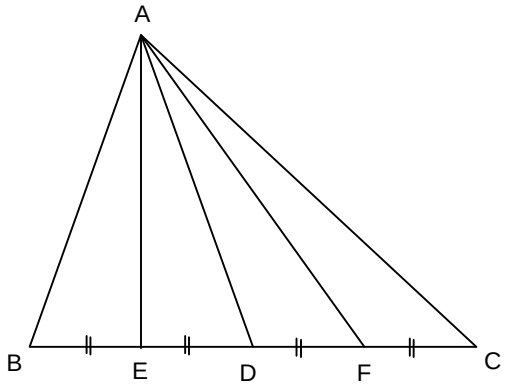
82. C

Sol. In ΔABC, $AD^2 = m_1^2 = \frac{c^2+b^2}{2} - \frac{a^2}{4}$

In ΔABD, $AE^2 = m_2^2 = \frac{AD^2+c^2}{2} - \frac{\left(\frac{a}{2}\right)^2}{4}$

[Apollonius Theorem]

In ΔADC, $AF^2 = m_3^2 = \frac{AD^2+b^2}{2} - \frac{\left(\frac{a}{2}\right)^2}{4}$



$$\therefore \quad m_2^2 + m_3^2 = AD^2 + \frac{b^2+c^2}{2} - \frac{a^2}{8}$$

$$= m_1^2 + \frac{a^2}{4} - \frac{a^2}{8} = 2m_1^2 + \frac{a^2}{8}$$

$$\Rightarrow \quad m_2^2 + m_3^2 - 2m_1^2 = \frac{a^2}{8}$$

83. B

Sol. We know that $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$

$$\Rightarrow \quad |\text{adj}(\text{adj}(\text{adj } A))| = |\text{adj } A|^{(n-1)^2}$$

$$= |A|^{(n-1)^3} = 2^8 = 256$$

84. B

Sol. The total number of third order determinants is 9! Since the number of determinants is even and in which there are 9!/2 pairs of determinants which are obtained by changing two consecutive rows,

So $\sum_{i=1}^{n} D_i = 0$.

85. C

Sol. $\left(A^{-1}BA\right)^2 = \left(A^{-1}BA\right)\left(A^{-1}BA\right)$

$= A^{-1}B\left(AA^{-1}\right)BA$

$\Rightarrow A^{-1}BIBA = A^{-1}B^2A$

$\left(A^{-1}BA\right)^3 = \left(A^{-1}B^2A\right)\left(A^{-1}BA\right)$

$= A^{-1}B^2\left(AA^{-1}\right)BA$

$= A^{-1}B^3A$ and so on

$\therefore \left(A^{-1}BA\right)^n = A^{-1}B^nA$

86. B

Sol. Let $f(x) = ax^2 + x + c - a$

$f(1) = c + 1 > 0 \quad (\because c > -1)$

$\therefore$ Given expression is positive for every $x \in R$

So, $f\left(\frac{1}{2}\right) > 0 \Rightarrow \frac{a}{4} + \frac{1}{2} + c - a > 0 \Rightarrow 4c - 3a + 2 > 0$

$\Rightarrow 4c + 2 > 3a.$

87. B

Sol. Let α be a common root then $\alpha^3 + 2a\alpha + 2 = 0$ and $\alpha^4 + 2a\alpha^2 + 1 = 0$

i.e. $\alpha^4 + 2a\alpha^2 + 2\alpha = 0$ ....(i)

$\alpha^4 + 2a\alpha^2 + 1 = 0$ ...(ii)

From (i) and (ii) $2\alpha - 1 = 0 \Rightarrow \alpha = \frac{1}{2}$

So, $\left(\frac{1}{2}\right)^3 + 2a\frac{1}{2} + 2 = 0 \Rightarrow \frac{1}{8} + a + 2 = 0 \Rightarrow a = -2 - \frac{1}{8} = -\frac{17}{8}$.

88. B

Sol. First, 6 distinct digits can be selected in ${}^{10}C_6$ ways. Now the position of smallest digit in them is fixed i.e. position 4. Of the remaining 5 digits, two digits can be selected in ${}^5C_2$ ways. These two digits can be placed to the right of 4th position in one way only. The remaining three digits to the left of 4th position are in the required order automatically.
So $n(S) = {}^{10}C_6 \times {}^5C_2 = 210 \times 10 = 2100$.

89. C

Sol. Clearly $p = 6! \, {}^{7}P_4$

$q = 7! \, . \, 4!$

$\frac{p}{q} = 5$

90. C

Sol. When 4 points are selected we get one intersecting point. So probability is $\frac{{}^{n}C_4}{({}^{n}C_2 - n)C_2}$.